

Leakage-Safe Reproduction of EvoC2Rust with CodeWeaver

Abstract

We reproduce the publicly executable portion of EvoC2Rust on the paper's disclosed Vivo-Bench repository

Experimental coverage and artifact availability

RQ	Surface	Status	Reason
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RQ1	Vivo-Bench ICompRate	measured	cumulative replacement with C fallback
RQ1	Vivo-Bench AccRate-P/R	unavailable	human-corrected Rust references are unreleased
RQ1	Vivo-Bench SafeRate	measured	candidate production Rust only
RQ2	Vivo-Bench FCompRate/TestRate	measured	19 modules and 125 active pinned tests
RQ1/RQ2	C2R-Bench	unavailable	C2R-Bench sources, tests, and corrected Rust references are unreleased
RQ3	EvoC2Rust ablations	reference_only	implementation, feature mappings, repairs, and C2R-Bench
RQ4	scale and time figures	reference_only	requires unreleased C2R projects and EvoC2Rust runtime

Table 4 extension: project translation

Dataset	Model	Method	ICompRate	AccRate-P	AccRate-R	SafeRate
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Vivo-Bench	DeepSeek-V3	EvoC2Rust	100.00%	99.83%	99.86%	98.00%
Vivo-Bench	Qwen3-32B	EvoC2Rust	87.65%	87.92%	83.45%	98.22%
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver repetition 1	100.00%	N/A	N/A
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver repetition 2	100.00%	N/A	N/A
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver repetition 3	100.00%	N/A	N/A
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver mean	100.00%	N/A	N/A

Table 5 extension: module translation

Dataset	Model	System/run	FCompRate	TestRate
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Vivo-Bench	DeepSeek-V3	19 projects	99.07%	98.50%
Vivo-Bench	Qwen3-32B	19 projects	87.65%	84.57%
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver repetition 1	100.00%
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver repetition 2	100.00%
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver repetition 3	100.00%
Vivo-Bench (pinned revision)		gpt-5.6-sol	CodeWeaver mean	100.00%

Exact CodeWeaver repetitions

Rep	Terminal	Integrity	IComp	FComp	Tests	TestRate	SafeRate	Elapsed
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1	15/15	15/15	19/19 (100.00%)	19/19 (100.00%)	125/125	100.00%	31.21%	11.22 h
2	15/15	15/15	19/19 (100.00%)	19/19 (100.00%)	125/125	100.00%	35.50%	12.03 h
3	15/15	15/15	19/19 (100.00%)	19/19 (100.00%)	125/125	100.00%	26.07%	10.86 h

Methodology

The benchmark is pinned at AtomGit commit ``c88cef1a1d15079478be14ab361dda8f3b49fee2``. C2Rust 0.22.1 deri

Threats to validity

The paper reports 113 Vivo-Bench test cases, while the pinned public revision enables 125 test functions

Reference-only experiments

The complete published Table 4, C2R-Bench portion of Table 5, and Table 6 ablation values are preserved

Provenance

- Paper DOI: ``10.1145/3786583.3786856``
- Paper version: ``2508.04295v4``
- Public benchmark: ``https://atomgit.com/vivoblueos2/vivo_blueos_contest2_c2rust`` at ``c88cef1a1d15079478``
- Protocol: ``{"agent_timeout_seconds": 5000, "effort": "max", "evaluated_repetitions": 3, "max_iter": 5,``
- Evaluation provenance: ``{"cargo": {"returncode": 0, "status": "measured", "value": "cargo 1.92.0-night``